

AMIRHESAM GHAHARI

Senior Data Engineer

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WORK HISTORY

Senior Data Engineer

TELUS Health 📅 04/2022 - Present 📍 Toronto, Canada

Spearheaded the design, implementation, maintenance, and optimization of **cloud-native data pipelines**, significantly improving data accuracy, latency and workflow efficiency.

- Architected scalable data models using **DBT**, **Python** and **Airflow** implementing **dimensional** and **star schema** best practices, enhancing data accessibility for analytics and reporting teams by **50%**
- Employed **Git workflows**, **Docker** and **Kubernetes** to automate ETL processes, reducing manual intervention and improving overall system reliability
- Revamped legacy ETL systems, deploying **real-time data processing frameworks** via **AWS DMS**, **Kinesis**, and **Lambda**, increasing data processing efficiency and reducing data latency by **80%**
- Established **multi-layered data architectures** using **AWS Lambda**, **Athena**, **RDS** and **Glue**, resulting in **40%** faster analytics reporting and improving data governance compliance
- Developed **optimized SQL queries** that improved reporting efficiency across multiple business intelligence tools such as **PowerBI** and **Tableau**, enabling data-driven decisions and reducing query execution times by **20%**
- Collaborated across teams to implement **innovative solutions**, improving operational efficiency, focusing on problem solving attitude, performing code reviews and resolving data integrity issues with creative solution.

Data Engineer

Data Sciences Inc. 📅 12/2020 - 04/2022 📍 Toronto, Canada

Developed and refined **advanced data pipelines**, ensuring data accuracy for high-profile projects with the Liberal Party of Canada and Ontario Liberal Party.

- Exploited **SQL**, **NoSQL databases** to process datasets in data warehouse, and improve cleaning and processing times by **30%** while ensuring compliance with data governance policies
- Systematized workflows using **custom Python scripts**, reducing manual effort by **40%** and accelerating **data migration processes**
- Designed and implemented **cloud-based pipelines** with **AWS** services such as **Lambda**, **Redshift**, **Step Function** and **EventBridge**, achieving cost efficiency and reducing deployment times
- Modernized **ETL workflows**, enabling **real-time data integration** pipelines providing data insights and improving stakeholder reporting turnaround
- Brainstormed with researchers to align data systems with project objectives, to ensuring accuracy, consistency, and adherence to **data governance** and **analytical standards**, supporting strategic decision-making

Health Project Data Engineer

Dahdaleh Institute for Global Health Research

📅 09/2019 - 09/2020 📍 Toronto, Canada

Formulated and launched a **data-driven decision support system**, leveraging scenario modelling and predictive analytics to assess the impacts of global warming on health factors, driving actionable insights for research and policy decisions

- Acquired and managed large, complex datasets from diverse sources, utilizing a **MongoDB-supported database** to ensure scalable, efficient data storage and seamless integration for advanced research and analytics
- Conducted comprehensive data cleaning, preprocessing, and exploration using **Python libraries** (e.g., **Pandas**, **NumPy**, **Matplotlib**), laying the foundation for advanced analysis and delivering critical research insights
- Enhanced ETL workflows by re-engineering database structures and ETL pipelines, significantly enhancing **data processing performance** by **50%** and ensuring high-quality, reliable outputs for **critical analytical tasks**

SUMMARY

Dynamic and results-oriented Senior Data Engineer with proven expertise in designing, implementing, and optimizing **ETL pipelines**, **data workflows**, and **big data solutions**. Proficient in **Python** and **SQL**, with advanced knowledge of **AWS services** (e.g., **S3**, **Lambda**, **RDS**, **Redshift**, **Kinesis**, **Glue**, **Athena**) to architect scalable, resilient cloud-based data systems. Skilled in data modelling (**star** and **dimensional schema**), orchestration tools like **Apache Airflow**, and big data technologies such as **Apache Spark** and **Hadoop**. Delivered measurable improvements in ETL processing times, cost savings on cloud resources, and query performance. Adept at applying **CI/CD** practices using **Jenkins**, **Git workflows**, **Docker** and **Kubernetes** to streamline deployments, enhance efficiency, and ensure system reliability. Recognized for implementing innovative, data-driven solutions that improve operational decision-making and enable actionable insights.

EDUCATION

Bachelor of Science Computer Science

York University

📅 09/2018 - 08/2020 📍 Toronto, Canada

Master of Science Systems Engineering

Amirkabir University of Technology

📅 09/2013 - 01/2016 📍 Tehran, Iran

Bachelor of Science Industrial Engineering

Amirkabir University of Technology

📅 09/2009 - 06/2013 📍 Tehran, Iran

SKILLS

Programming Languages:

Python, JavaScript, Java

Databases:

PostgreSQL, MySQL, MongoDB, DynamoDB

Cloud Platforms:

AWS S3, Redshift, RDS, Lambda, DMS, Kinesis, DMS, Glue, EMR, Athena, DynamoDB

Data Modelling:

Dimensional Modelling, Star & Snowflake Schema

Version Control and Containerization:

Git, GitHub, Docker, Kubernetes

Data Engineering Tools:

Apache Spark, Hadoop, Airflow, Kafka